SMART GRID GUARD

IoT-Based Power Monitoring System

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1.PROBLEM STATEMENT

Electrical power systems often experience voltage fluctuations and unexpected faults, but users are usually unaware of these issues until they cause equipment damage or power interruptions. The lack of continuous monitoring and instant alert mechanisms makes it difficult to identify and respond to problems at an early stage.

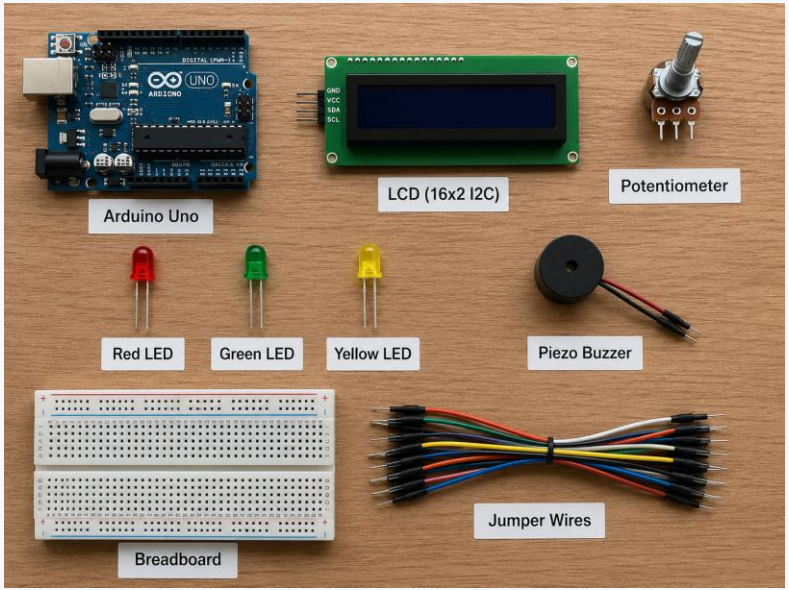
Smart Grid Guard was developed to solve this problem by continuously monitoring voltage conditions and providing real-time alerts whenever abnormal situations are detected.

2. PROPOSED SOLUTION

Smart Grid Guard is a smart power monitoring system that continuously observes voltage conditions and detects abnormal situations in real time. It provides visual alerts through LEDs, displays system status on an LCD screen, activates a buzzer during critical conditions, and sends Telegram notifications to users. This helps identify potential problems early and improves the reliability of power systems.

3.COMPONENTS USED

1. Arduino Uno
2. 16×2 I2C LCD Display
3. Potentiometer
4. Green, Yellow & Red LEDs
5. Piezo Buzzer
6. Bread Board
7. Jumper Wires



Component	Why It Is Used
1.Arduino Uno	Controls the entire system
2.LCD (16×2 I2C)	Displays system status
3.Potentiometer	Simulates voltage changes
4.Green LED	Indicates normal condition
5.Yellow LED	Indicates warning condition
6.Red LED	Indicates fault condition
7.Piezo Buzzer	Provides sound alerts
8.Breadboard	Builds circuit without soldering
9.Jumper Wires	Connects all components

4.IMPLEMENTATION

The implementation of **Smart Grid Guard** was carried out using Arduino Uno, a 16×2 I2C LCD display, LEDs, a piezo buzzer, and a potentiometer in the Tinkercad simulation environment.

Initially, all hardware components were connected to the Arduino according to the circuit design. The potentiometer was connected to an analog input pin to simulate different voltage levels. The LCD display was interfaced using I2C communication to display system status. Three LEDs (Green, Yellow, and Red) and a piezo buzzer were connected as output devices for visual and audible alerts.

Based on the voltage level, the system operates in three different states:

- **Normal Condition:** Green LED glows and the LCD displays a normal status message.
- **Warning Condition:** Yellow LED glows and the LCD displays a warning message.
- **Critical Fault Condition:** Red LED and buzzer are activated, and the LCD displays a fault alert.

To enable remote monitoring, a Python script was developed and executed through the command prompt. The script receives data from the Arduino via serial communication and forwards the system status to a Telegram Bot. Whenever a warning or fault condition is detected, an instant notification is sent to the user through Telegram.

The complete system was tested in Tinkercad by varying the potentiometer values and observing the corresponding changes in LEDs, buzzer, LCD messages, and Telegram notifications.

FLOW :

Potentiometer

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Arduino Uno

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Condition Analysis

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LCD Display
LED Indicators
Buzzer Alert

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Python Script

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Telegram Bot

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User Notification

TINKERCAD CIRCUIT

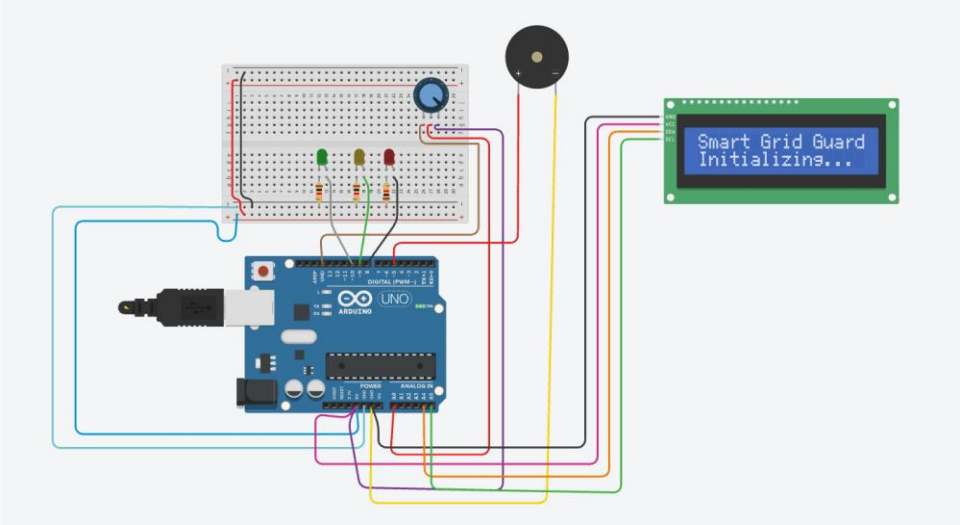
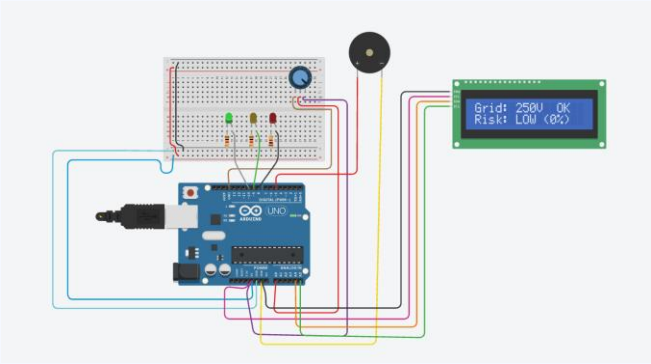


Figure: Complete Smart Grid Guard circuit designed and tested in Tinkercad simulation environment

RESULTS ACHIEVED

- Displayed real-time system status on LCD.
- Generated visual alerts using LEDs.
- Activated buzzer during critical conditions.
- Sent Telegram notifications successfully.
- Tested under multiple voltage conditions in Tinkercad.
- Successfully monitored voltage variations.



CONCLUSION

Smart Grid Guard successfully demonstrates how embedded systems and IoT technologies can be used to create a smart and efficient power monitoring solution. The system continuously monitors voltage conditions, identifies abnormal situations, and provides instant alerts through LEDs, a buzzer, and Telegram notifications. By enabling both local and remote monitoring, the project helps improve reliability, reduce response time, and prevent potential power-related issues. This project serves as a practical step toward the development of smarter and more intelligent power management systems.

This project demonstrates a practical and cost-effective approach to real-time power monitoring and serves as a foundation for future smart-grid applications.

FUTURE SCOPE

Smart Grid Guard can be further enhanced by integrating advanced technologies to make the system more intelligent, scalable, and suitable for real-world power monitoring applications

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Future Enhancements

- **IoT Integration using ESP32/ESP8266**
  Enable direct internet connectivity without requiring a computer.
- **Cloud-Based Monitoring**
  Store and monitor data on cloud platforms for remote access from anywhere.
- **Mobile Application**
  Develop a dedicated mobile app for real-time monitoring and alerts.
- **Data Logging and Analytics**
  Record voltage data for analysis, reporting, and performance evaluation.
- **AI-Based Fault Prediction**
  Use machine learning algorithms to predict faults before they occur.
- **Smart Grid Integration**
  Integrate with larger smart-grid infrastructures for city-wide monitoring.
- **Multi-Parameter Monitoring**
  Monitor additional parameters such as current, power consumption, frequency, and temperature.
- **Automatic Fault Response**
  Automatically disconnect or protect equipment when critical conditions are detected.

THANK YOU!